IMPERIAL

# Department of Chemical Engineering

# Machine Learning for Chemical Engineering

Lecture 1a – Course Overview

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# Machine Learning for Chemical Engineering

Lecturer:

- Dr. Antonio del Rio-Chanona
- Optimisation and Machine Learning for Process Systems Engineering Group:

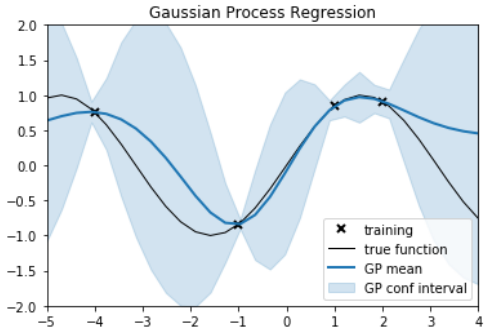
  https://www.optimlpse.co.uk/people/

In this course you will learn **Machine Learning and Data Science** with its **application to Chemical Engineering**.

Further material and resources will be highlighted in each section for a more comprehensive dive into the subject.

OPTIMLPSE

# What will you learn?



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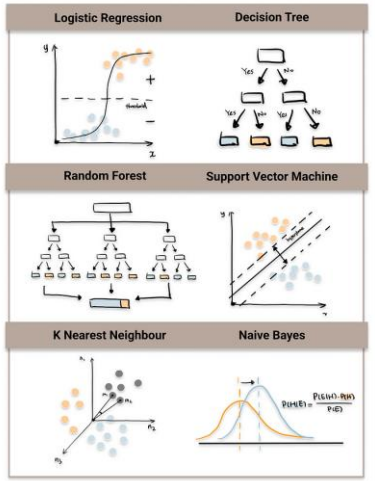
# What will you learn?

## 1. What is Machine Learning?

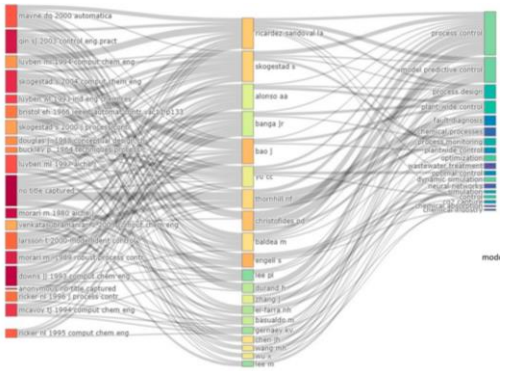
## 2. Machine Learning Models

**Supervised learning**



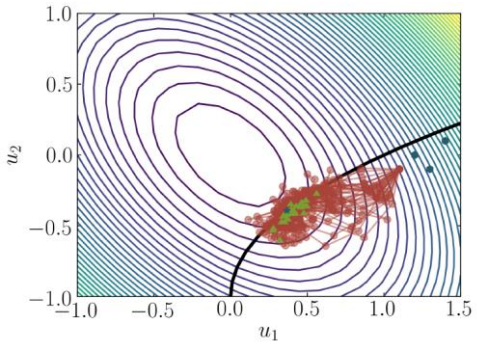

## 3. Data-driven fault and anomaly detection

**Unsupervised learning**



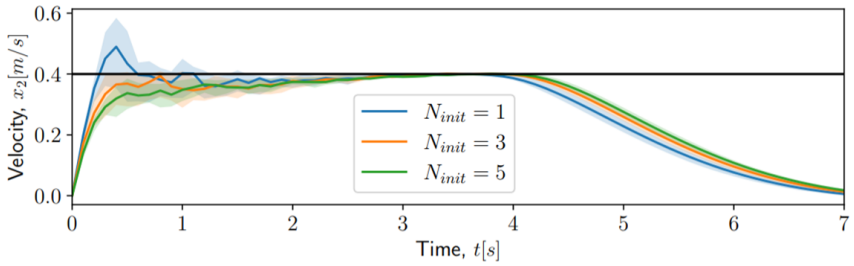
## 4. Data-Driven Optimization

**Bayesian optimization / Derivative free optimization**

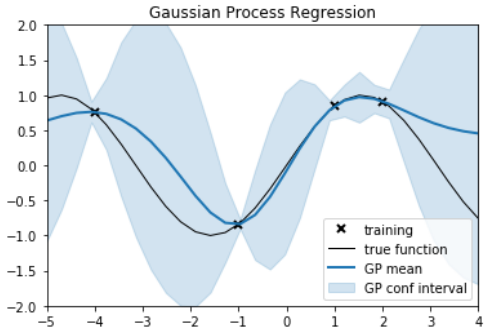



## 5. Reinforcement Learning and Data-Driven Control

**Reinforcement learning & time-series prediction**



# What is Artificial Intelligence and Machine Learning?



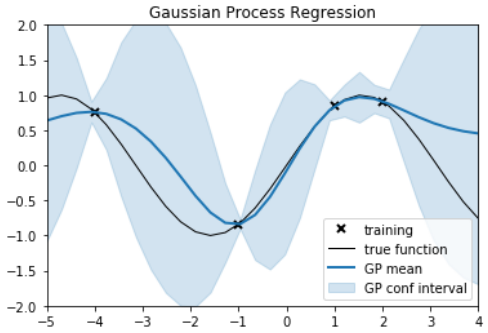
# Artificial Intelligence and Machine Learning?

- AI is the broader field of creating “machines” that simulate human-like intelligence, including human cognition, reasoning, problem-solving, language understanding, and perception.

- Machine learning is a subset of AI focused on developing algorithms that allow machines to learn and improve from data without being explicitly programmed for each specific task.

- Here we will focus on ML.

# What is Machine Learning?



# What is Machine Learning?

**“Traditional” computer science**

- Write exact instructions for a computer to do a specific task.
- If playing Chess:
  - If the opponent moves their horse move your bishop, else move your queen.
- Predicting cats and dogs from images:
  - If it has a pointed snout or is big, it is a dog, else it is a cat.

**Machine Learning**

- Provide examples to the machine.
- Machine learns to accomplish tasks based on the examples.

# What is Machine Learning?

**“Traditional” computer science**

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**Definition of Machine Learning (One of many)**

In 1959, Arthur Samuel: **“**field of study that gives computers the ability to learn without being explicitly programmed**”.**

# What is Machine Learning? – Three Categories

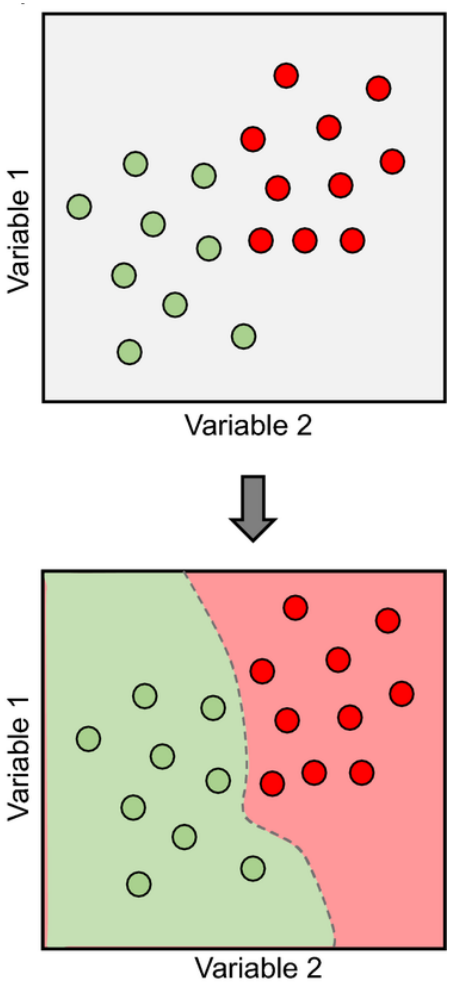
**Supervised Learning**

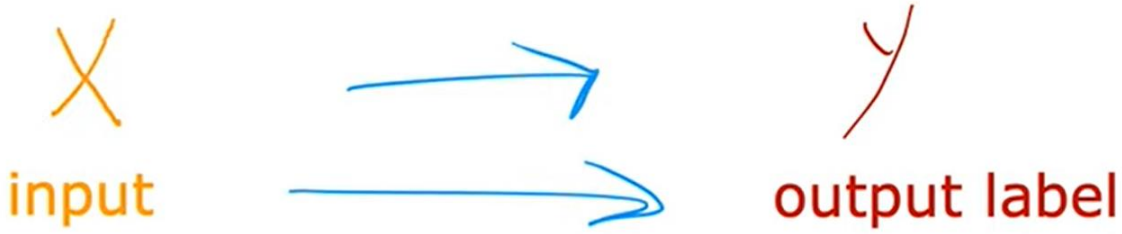
**Unsupervised Learning**

**Reinforcement Learning**

# What is Machine Learning? – Supervised Learning

## Supervised Learning




Learns from being given "right answers"

**Dog**

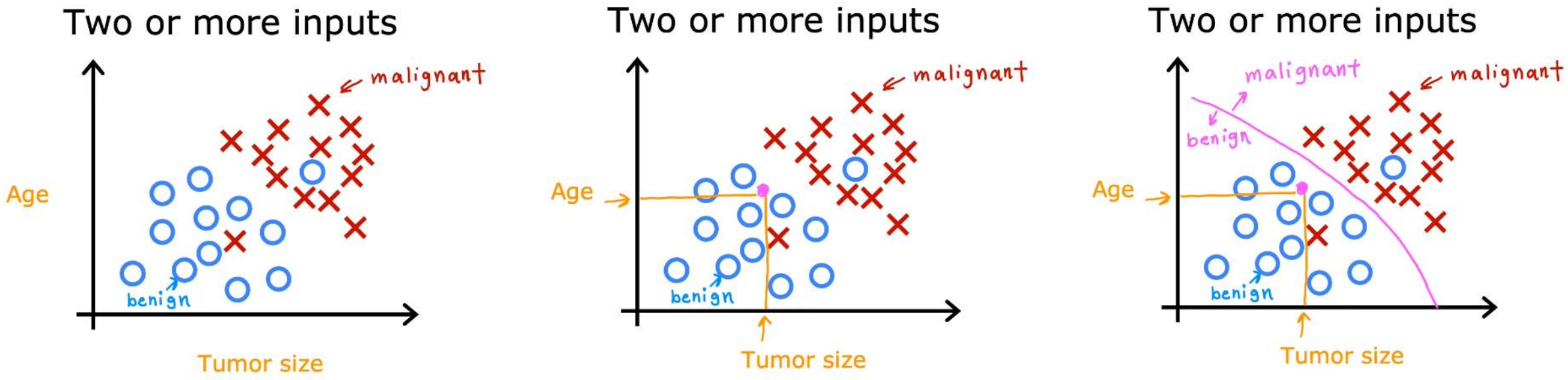
**Cat**

www.educba.com/what-is-supervised-learning/

# What is Machine Learning? – Supervised Learning

**Supervised Learning**

## Classification

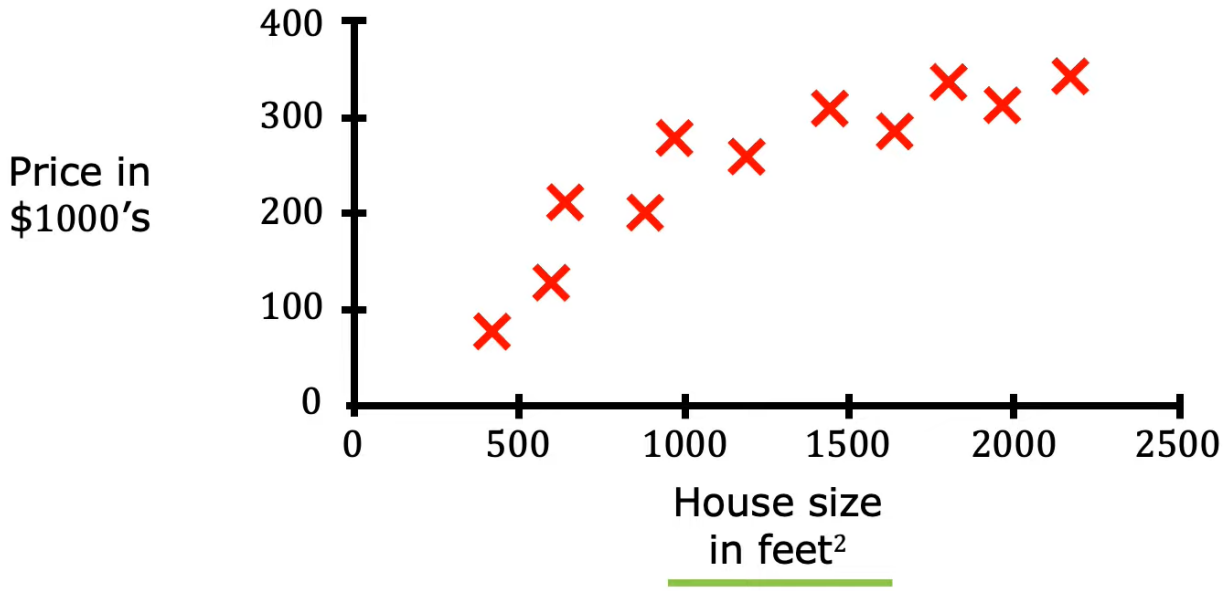


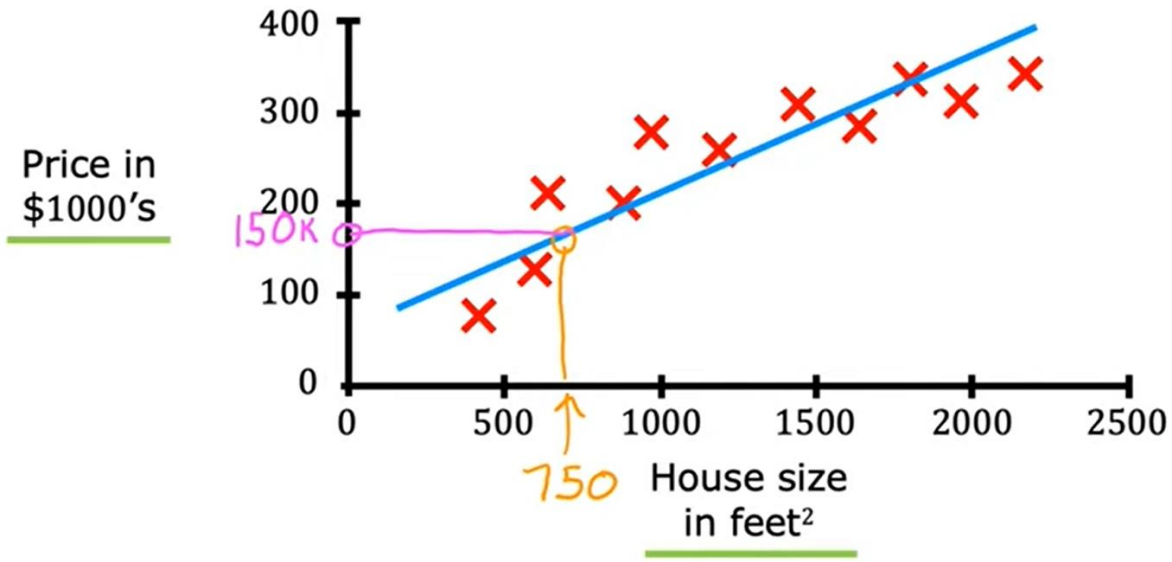

www.coursera.org/specializations/machine-learning-introduction

# What is Machine Learning? – Supervised Learning

**Supervised Learning**

## Regression

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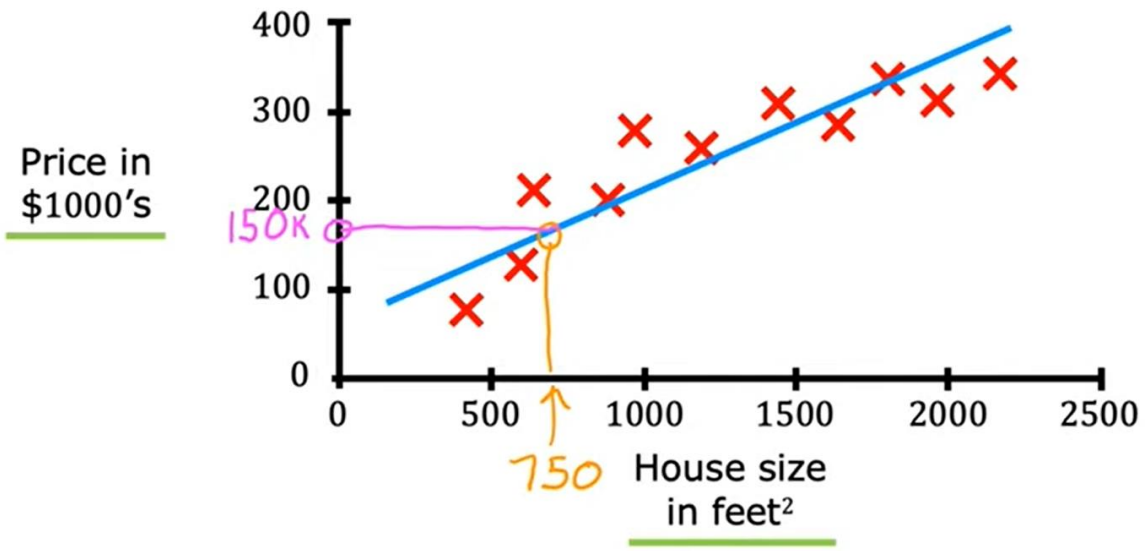
# What is Machine Learning? – Supervised Learning

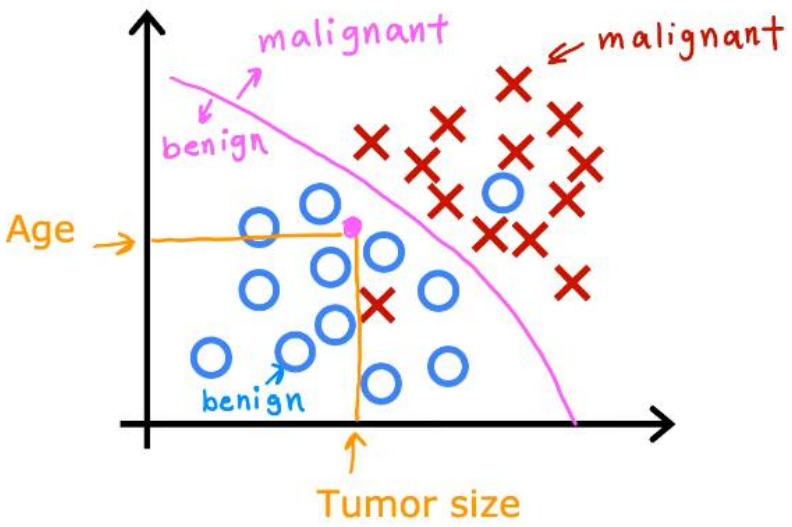
**Supervised Learning**

- Feeding the Learning agent with examples and trying to make it “generalize”.
- This can be thought as “**regression**” where the “machine” adjusts some parameters (or weights) to best capture the data.

**Regression**




**Classification**



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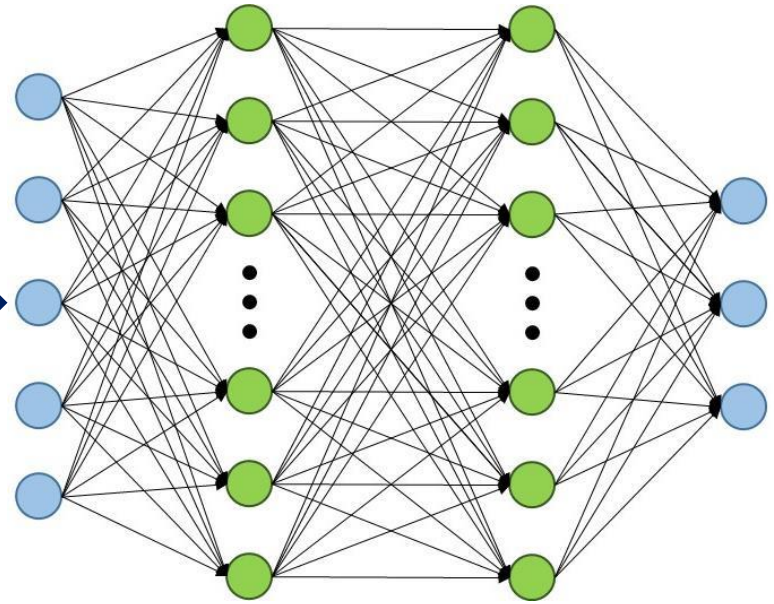
**Examples:**

**Train with many examples**



**cat, cat, cat ….**

**dog, dog, dog ….**

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**Supervised Learning**

- Feeding the Learning agent with examples and trying to make it “generalize”.
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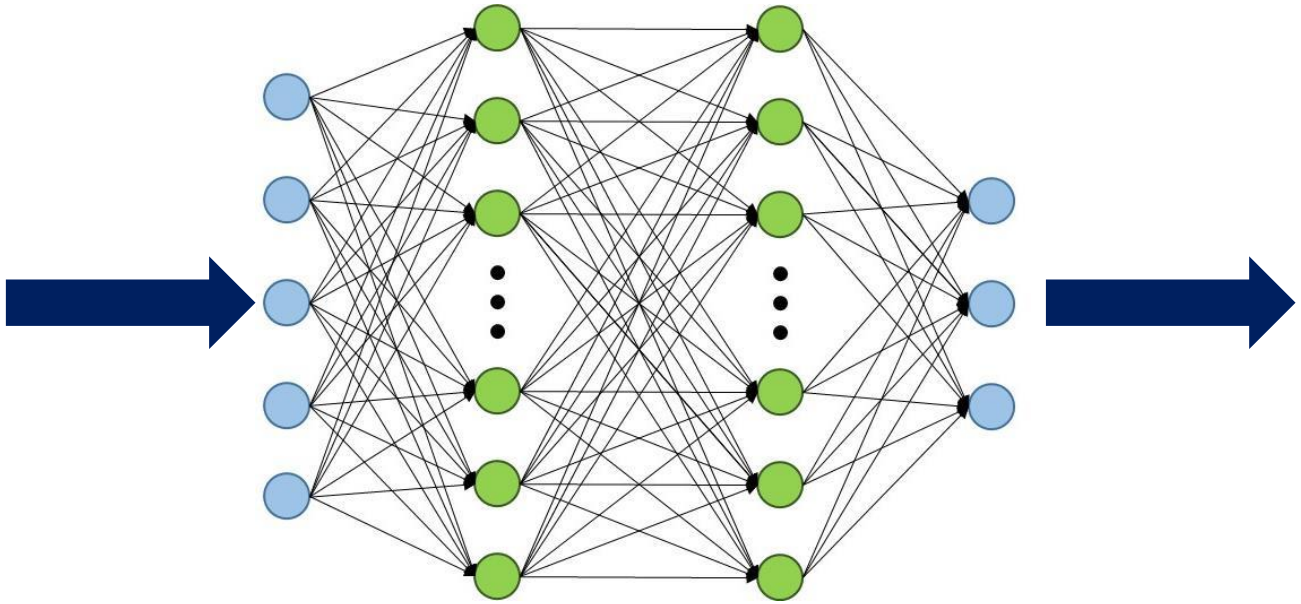
**Examples: Classification (**you predict classes, either a cat or a dog **– Boolean decisions)**

**New example**



**78% Dog**

**22% Cat**

# What is Machine Learning? – Supervised Learning

**Supervised Learning**

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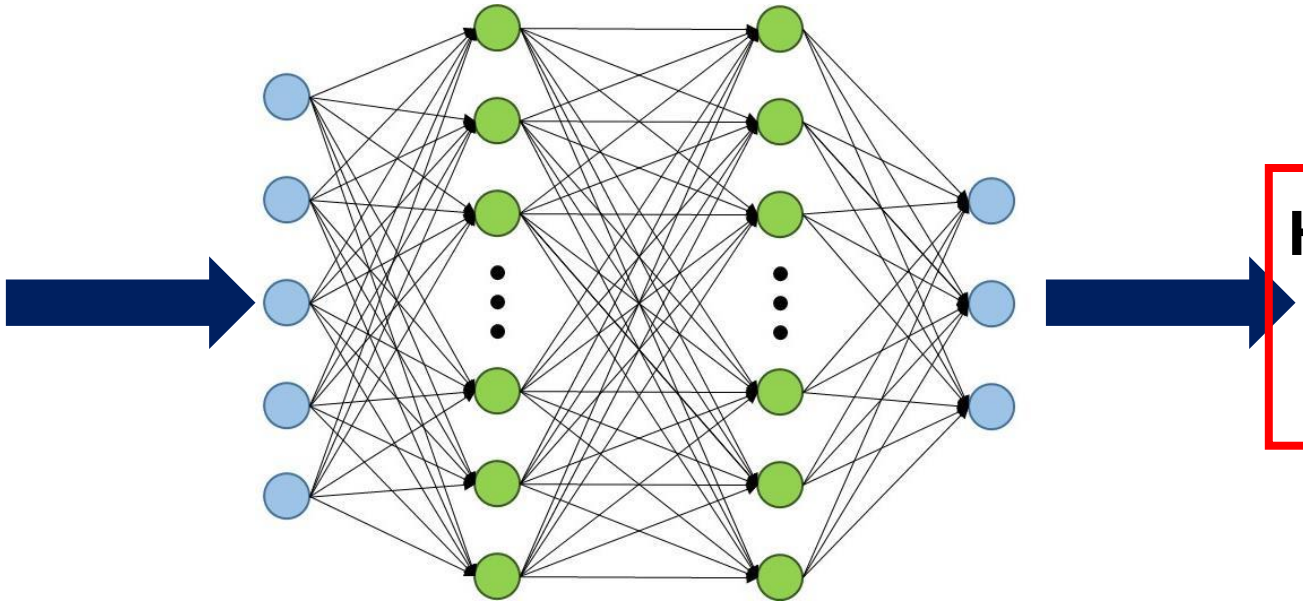
**Examples: Regression**

$m^2$
Postcode
# Windows



**Continuous value prediction**

**House Value**
**£££**

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**Supervised Learning**

- Feeding the Learning agent with examples and trying to make it “generalize”.
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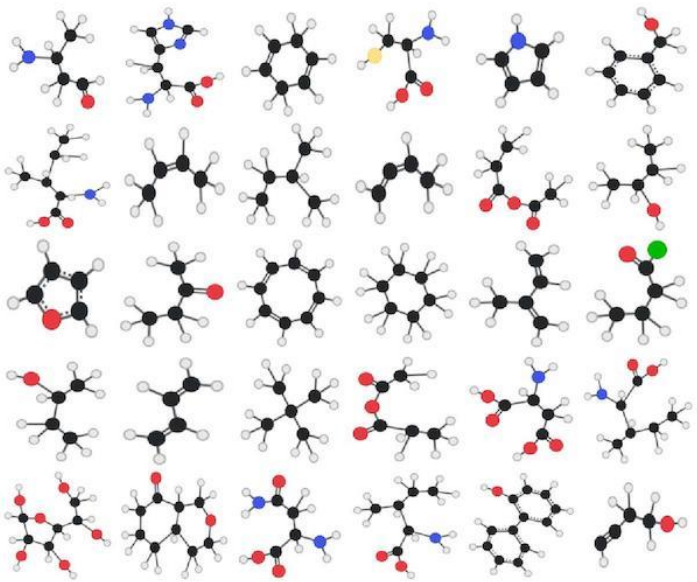
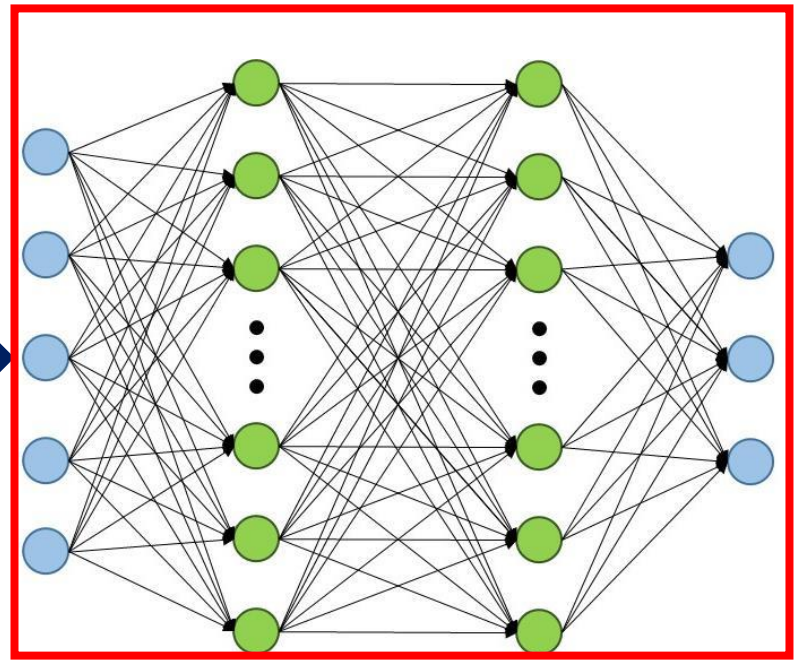
**Examples: Predicting molecular properties**

**This is a model!**

**(Some kind of approximation to your DFT model)**





**Dipole moments**

**Potential energy**

**Mulliken charges**

⋮

# What is Machine Learning? – Supervised Learning

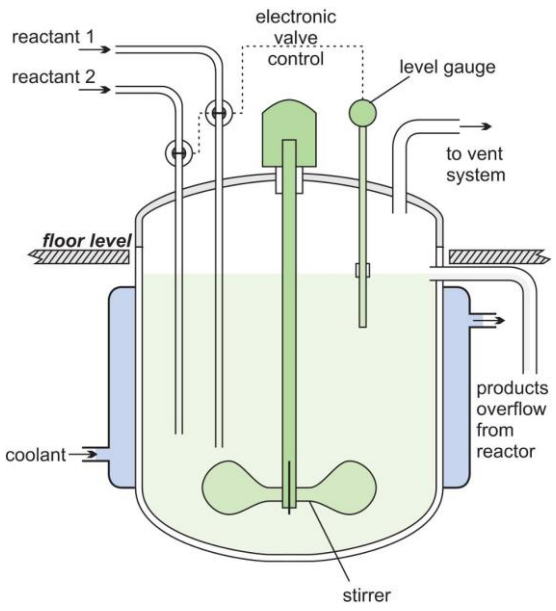
**Supervised Learning**

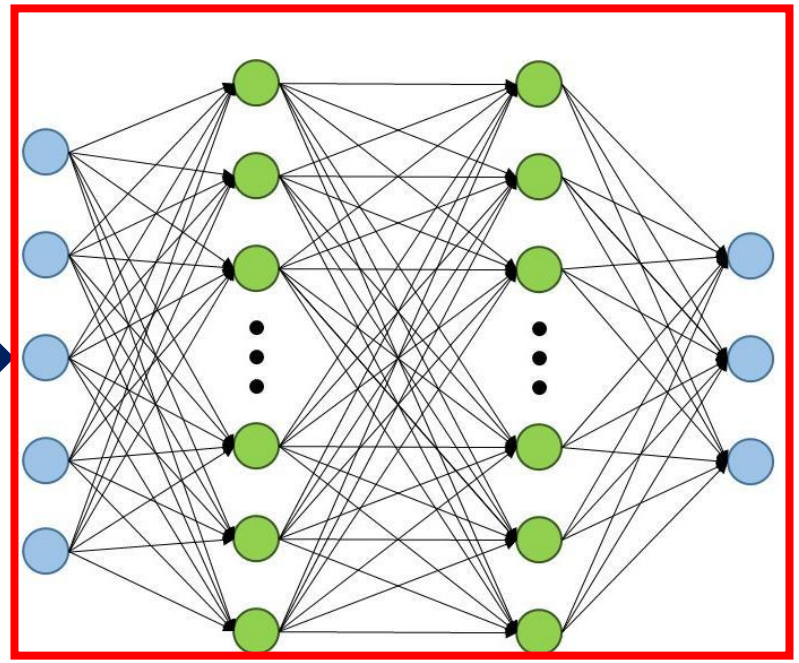
- Feeding the Learning agent with examples and trying to make it “generalize”.
- This can be thought as “**regression**” where the “machine” adjusts some parameters (or weights) to best capture the data.

**Examples: Modelling a chemical reactor**

**This is a model!**

**(Some kind of approximation to your Reactor model)**



**Concentrations at the outlet**

# What is Machine Learning? – Supervised Learning

**Supervised Learning**

**<u>Unsupervised Learning</u>**

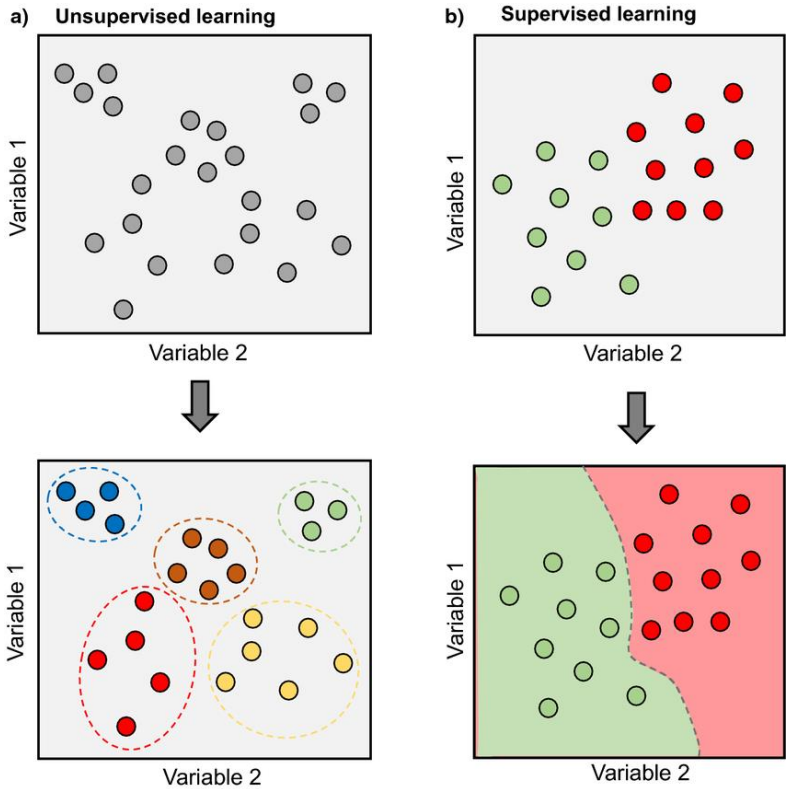
**Reinforcement Learning**

# What is Machine Learning? – Unsupervised Learning

**Unsupervised Learning**

- We do not provide ‘labels’ or ‘right and wrong answer’
- Give inputs, but no outputs
- Identify patterns or groups which remained previously unknown.

**<u>Example: Clustering</u>**



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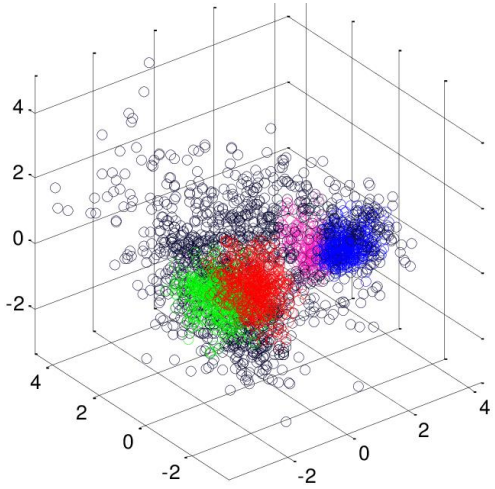


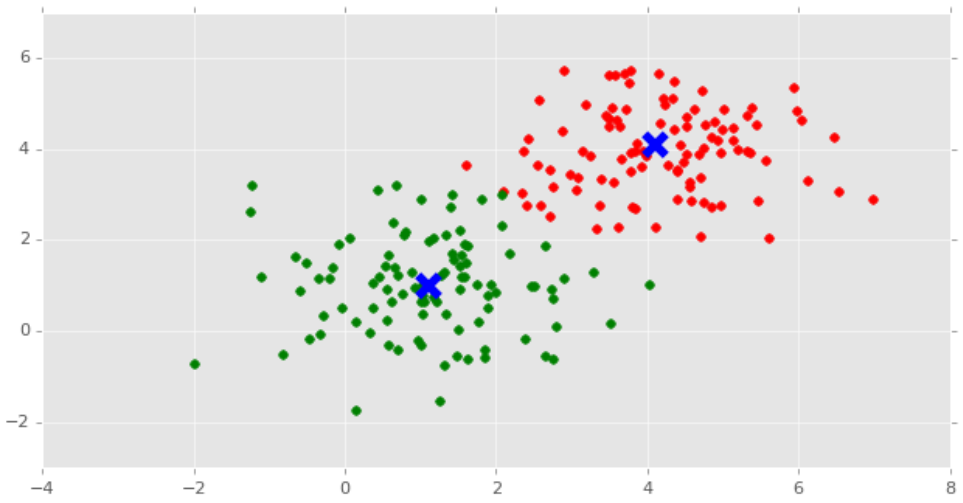

# What is Machine Learning? – Unsupervised Learning

**Unsupervised Learning**

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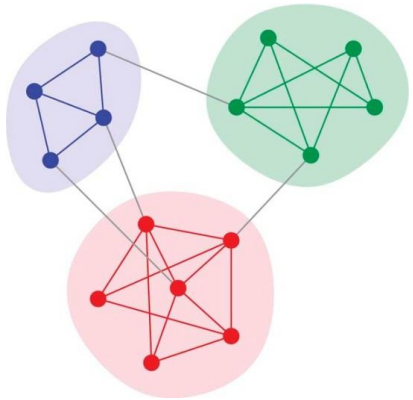
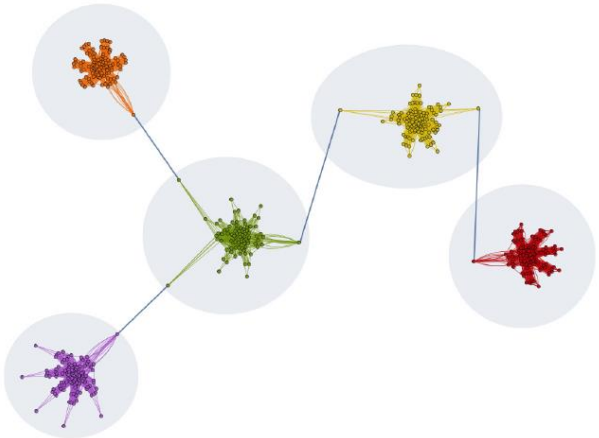
**<u>Example: Community detection</u>**





# What is Machine Learning? – Unsupervised Learning

**Unsupervised Learning**

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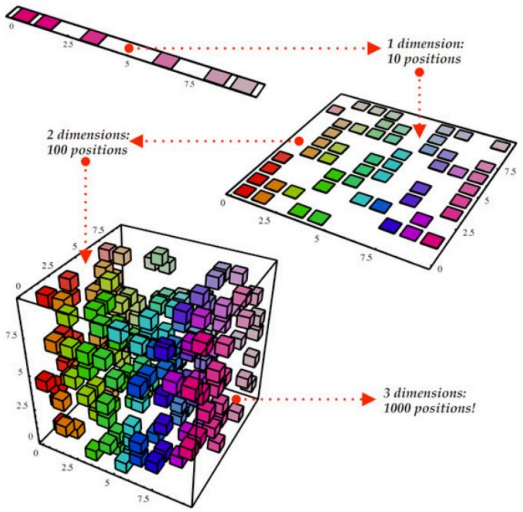
**<u>Example: Dimensionality Reduction</u>**



# What is Machine Learning? – Three Categories

**Supervised Learning**

**Unsupervised Learning**

**<u>Reinforcement Learning</u>**

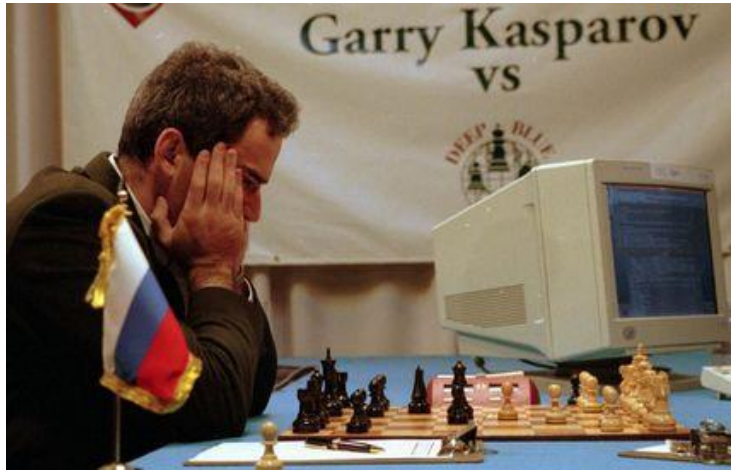
# What is Machine Learning? – Reinforcement Learning

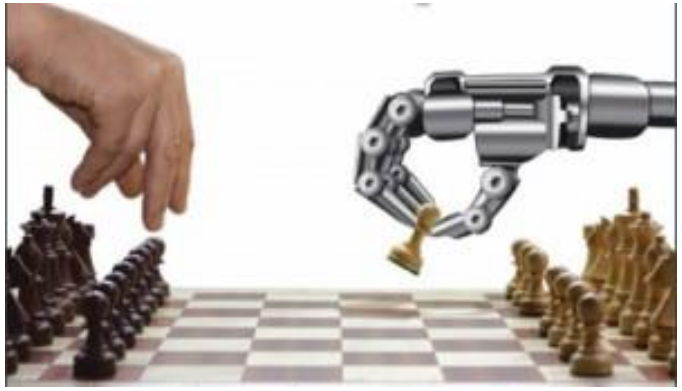
**Reinforcement Learning**

- Name comes from animal psychology, where we train animals/pets by reinforcing good behaviour, and discouraging bad behaviours.
- For Computers we give a positive “reward” for doing well (e.g. winning a game) and a give negative “reward” for doing bad (e.g. losing a game).

**<u>Example:</u>**

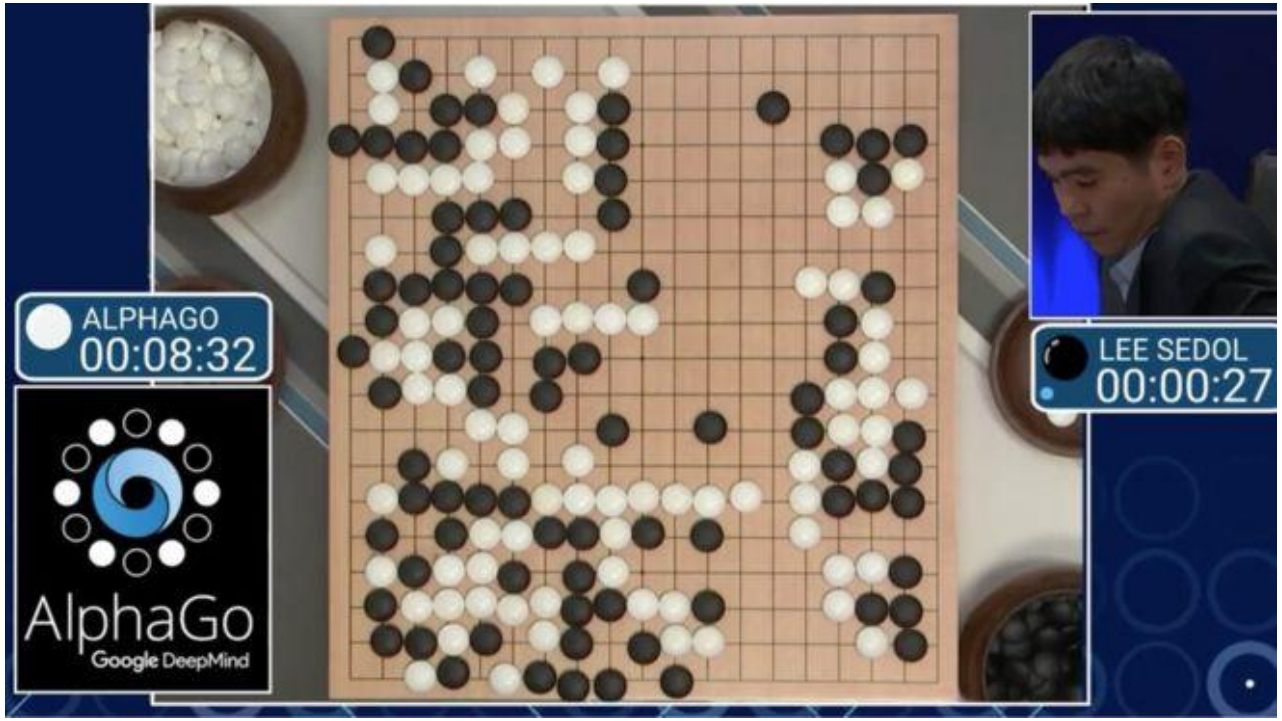
# Garry Kasparov vs Deep Blue (IBM)





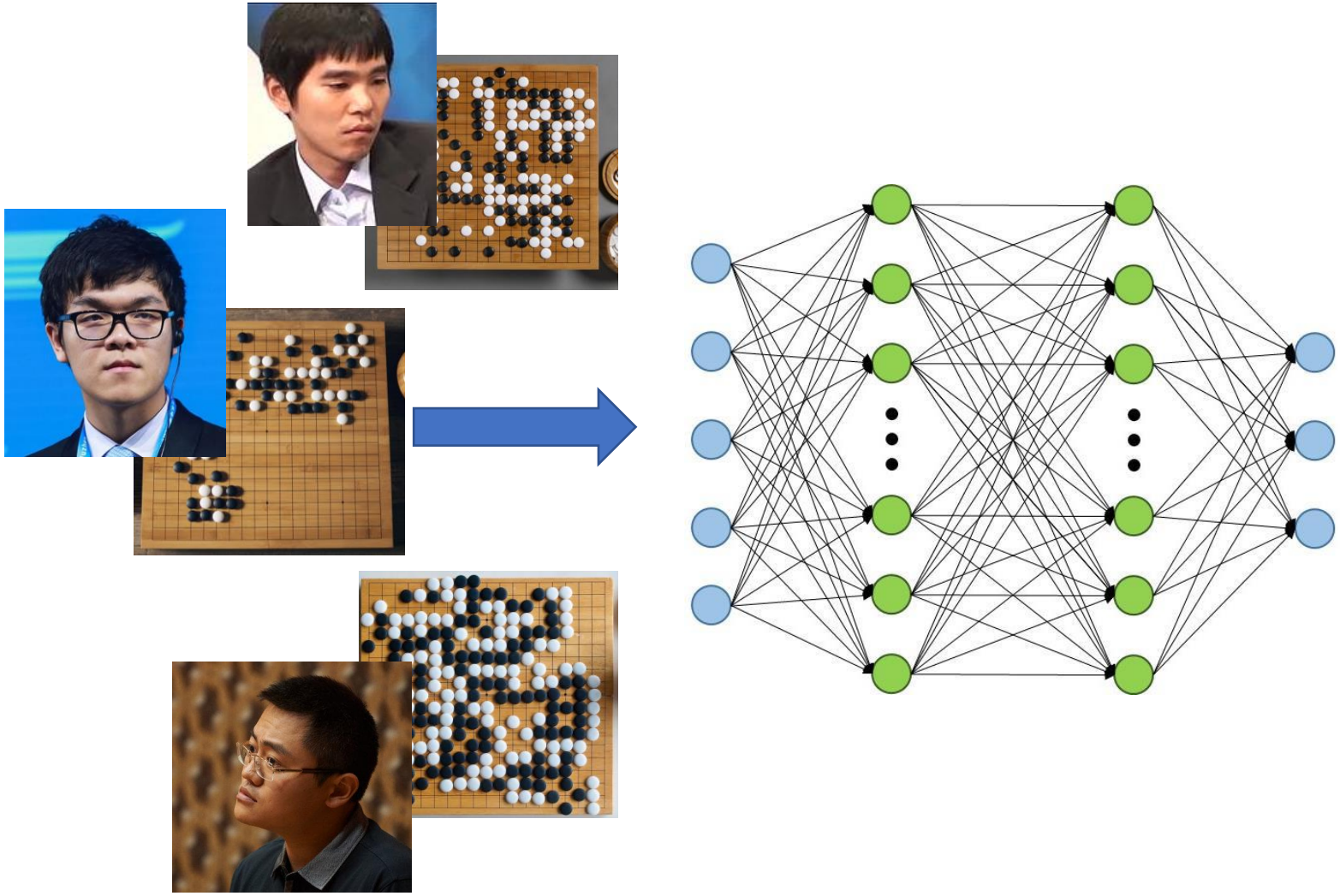
**Hardcoded by IBM engineers, with detailed instructions (i.e. if/else statements)**
**1-1 final score (1996-1997)**
**Can computers do better?**

# AlphaGO vs Lee Sedol



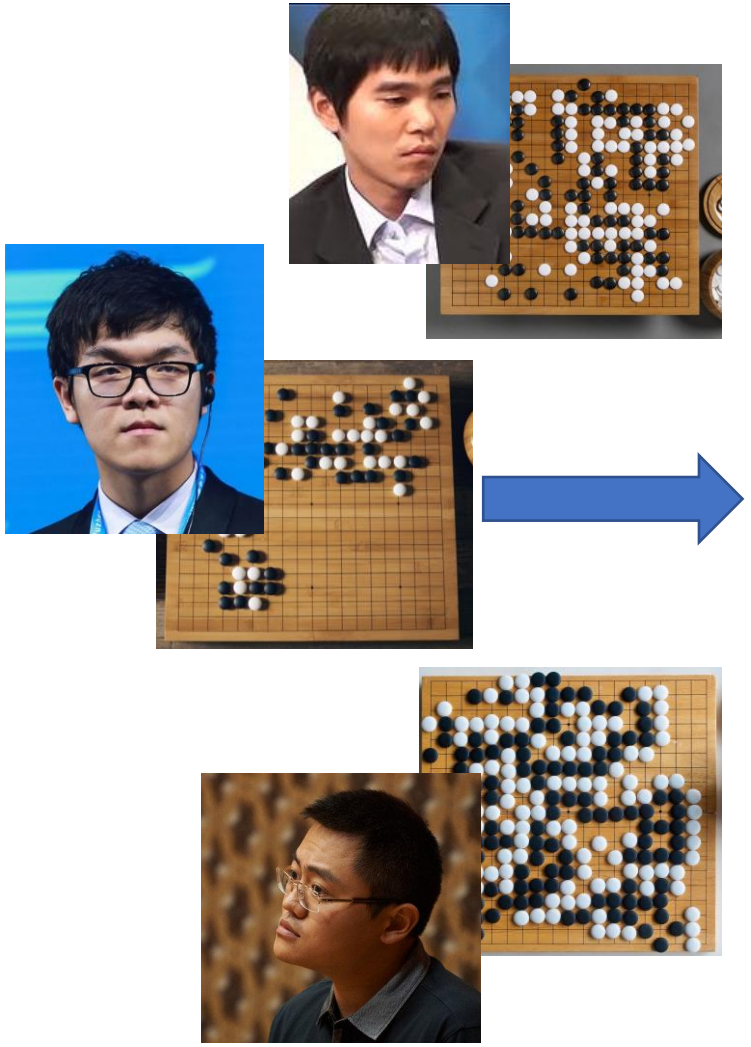

- **GO is MANY times more complex and with a broader search-space than chess.**
- **AlphaGo defeated Lee Sedol 4-1**
- **AlphaGo later defeated the No 1 in the world 3-0**
- **Later defeated five of the best players in the world combined**
- **AlphaGo ZERO defeated AlphaGO 100-0**

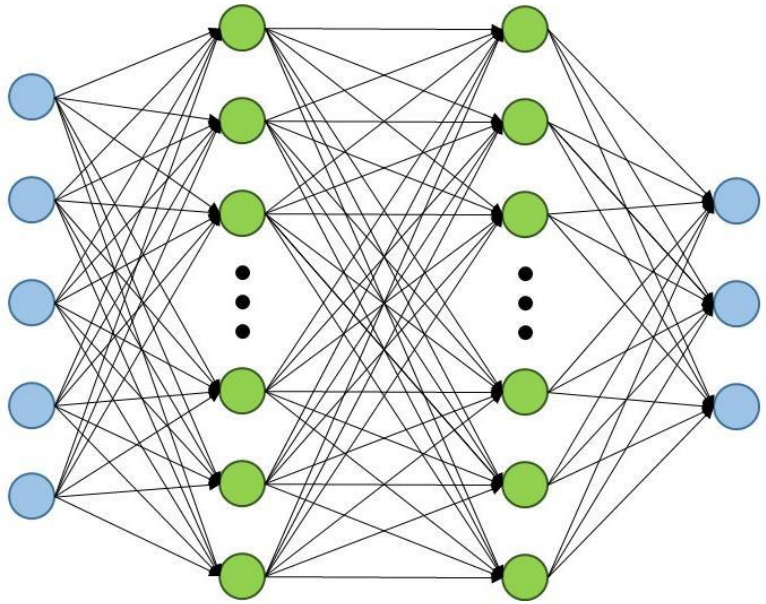
# Reinforcement Learning in action



***Apprenticeship learning*** *or* ***Behavioural cloning***

# Reinforcement Learning in action

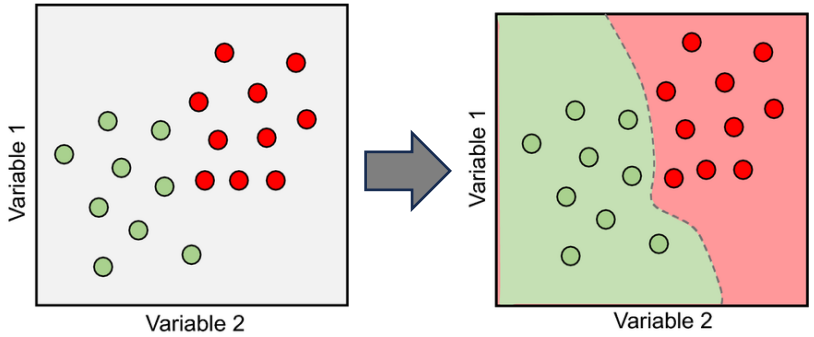




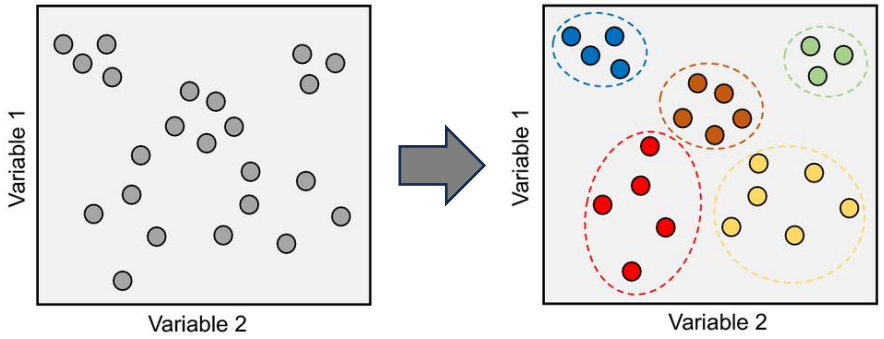
**Self-play**

# What is Machine Learning

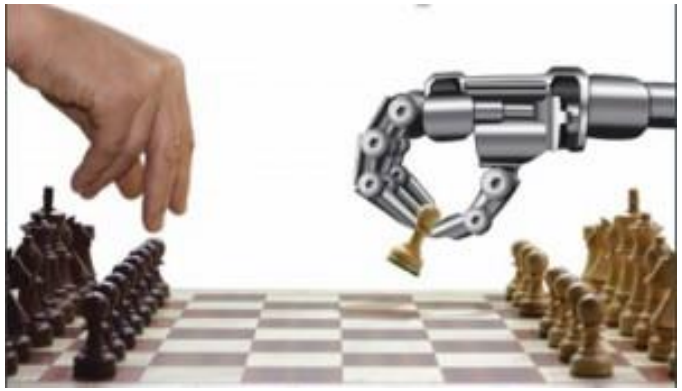
## Supervised Learning




## Unsupervised Learning




## Reinforcement Learning



www.educba.com/what-is-supervised-learning/

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# Department of Chemical Engineering

# Machine Learning for Chemical Engineering

## Lecture 1b – Supervised Learning: Linear Regression